

National Tsing Hua University
11320IEEM 513600
Deep Learning and Industrial Applications
Homework 4

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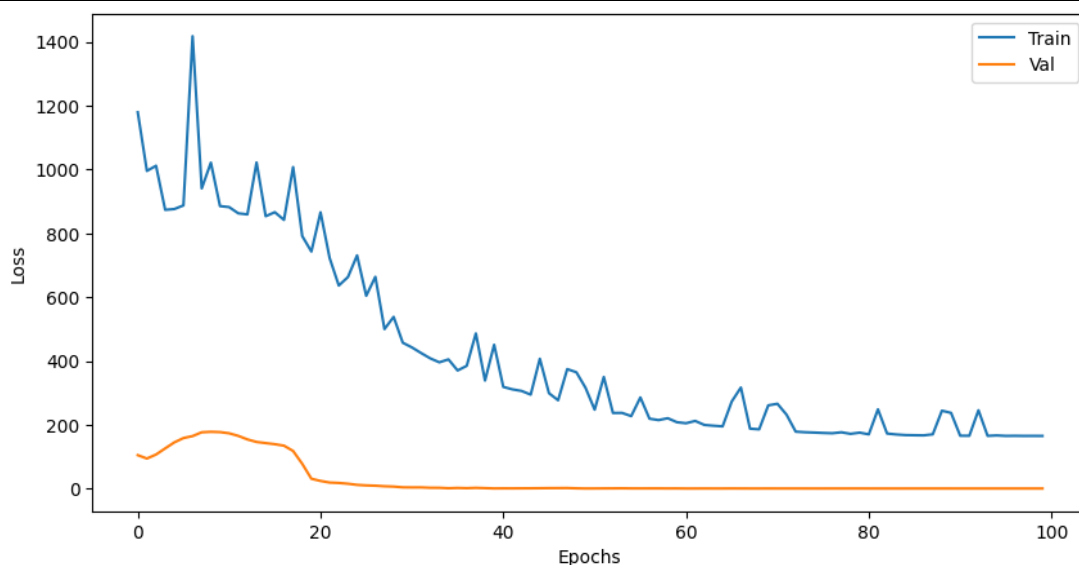
Student ID: 112034701

Due on 2025/05/01.

Note: DO NOT exceed 3 pages.

1. (15 points) Experiment with different window sizes and steps. Train the model using 3 different combinations of window size and step. Evaluate the Mean Squared Error (MSE) for each configuration. Report the MSEs using a table and analyze the results. (Approximately 100 words.)

Window size	step	Best val loss
10	15	246.9867
15	15	113.2701
15	20	0.5296



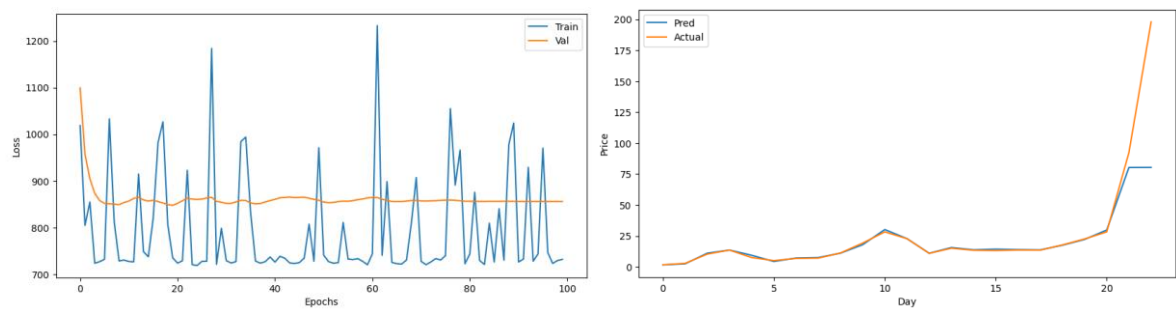
window size愈大代表可以看到的歷史資料愈多，而step愈大代表樣本比較稀疏，可以訓練得更快。根據實驗結果顯示，當組合為window size = 15 與step = 20的狀況，Best val loss最低，這個原因可能為比較大的window size提供了較多的歷史資訊，step的重疊度與樣本數量之間剛好取得平衡，提升模型的準確度。

2. (Approximately 200 words.)

- (i) (15 points) Include 'Volume' as an additional input feature in your model. Discuss the impact of incorporating 'Volume' on the model's performance.

將Volume加入input feature後，Best val loss由0.5296大幅上升至847.5449，顯示

模型預測能力明顯下降，推測可能為Volume的數值分布和價格變化的尺度差異過大，或是volume與high之間的關聯比較弱，導致noise增加。



- (ii) (15 points) Explore and report on the best combination of input features that yields the best MSE. Briefly describe the reasons of your attempts and analyze the final, optimal input combination.

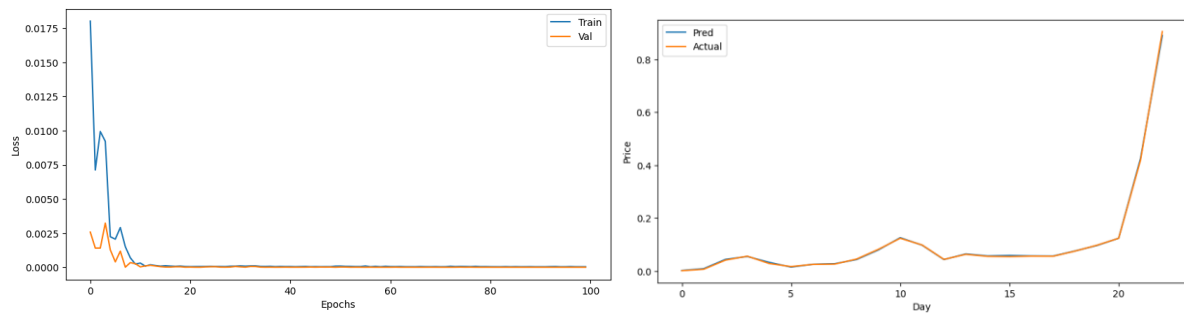
Combination	Best val loss	Reason
Open	431.2262	單一特徵測試
Open, Close	214.3081	當日開始與結束的價格變動差距
High, Low	360.5696	當日高低點差距
Open, Low, Close	396.1488	在較佳的 High, Low 組合下，加入當日起始
Open, Low, High	209.7000	在較佳的 High, Low 組合下，加入 High，但可能因為 label leakage 導致效果不佳
Open, Low, Close, High	115.9496	將 close 與 high 結合
Open, Low, Close, High, Volume	847.5449	特徵幾乎全選

根據上表，Open 單一特徵的資訊不足，而加入 Close 後能清楚當日趨勢，有顯著提升準確度。High Low 組合為當日價格區間，但沒有包含開盤收盤，因此 MSE 仍偏高。加入 High 的組合強化了模型的預測能力，而 Open Low Close High Volume 雖然最全面，但因為 Volume 特徵缺少標準化，導致訓練失衡。因此，本實驗最佳組合為 Open Low Close High，後續若針對 Volume 標準化，有機會成為最佳組合候選。

3. (15 points) Analyze the performance of the model with and without normalized inputs in Lab 4. You can use experimental results or external references (which must be cited) to support your conclusions on whether normalization improves the model's performance. (Approximately 100 words.)

根據實際 Normalization 後，在其餘參數不變的情況下，Open Low Close High Volume 組合的 Val loss 降為 0.000011，透過下方圖表可知，此模型在經過

Normalization後有很好的泛化結果，尤其是val loss比train loss (0.000047)的表現更佳，證明normalization在此問題中帶有非常正向的影響。



4. (10 points) Why should the window size be less than the step size in Lab 4? Do you think this is correct? If you use external sources, please include references to support your response. (Approximately 50 words.)

我認為window size比step size小是合理的設定，因為當window size比step size大時，會使相鄰時間序列樣本重疊度高，模型容易overfitting，而step size較大時可以讓樣本間距變遠，可以觀察到更多樣的行為模式。

5. (15 points) Describe one method for data augmentation specifically applicable to time-series data. Cite references to support your findings. (Approximately 100 words.)

在Time Series Data Augmentation for Deep Learning: A Survey這篇論文中，提出一個窗口變形 (window warping) 的技術，選擇性拉長或是收縮時間序列中的不同片段，通過對數據不同區域應用不同程度的變形效果，可以保留基本模型的真實變化，同時導入更重要的多樣性。另外此篇論文還提出窗口切片、抖動、縮放、旋轉、排列等合成數據生成技術。

Reference: Wen (2022), Time Series Data Augmentation for Deep Learning: A Survey

6. Discuss how to handle window size during inference in different model architectures (approximately 150 words):

- (i) (5 points) Convolution-based models: 通常需要固定的window size。
- (ii) (5 points) Recurrent-based models: Recurrent-based model有記憶能力，可以逐步接收時間序列資料，透過sliding window將新資料加進window並捨棄最舊的資料，或是用固定長度的window。
- (iii) (5 points) Transformer-based models: 通常仍會使用固定長度window來提升效率，長序列推論可搭配caching或sliding window。